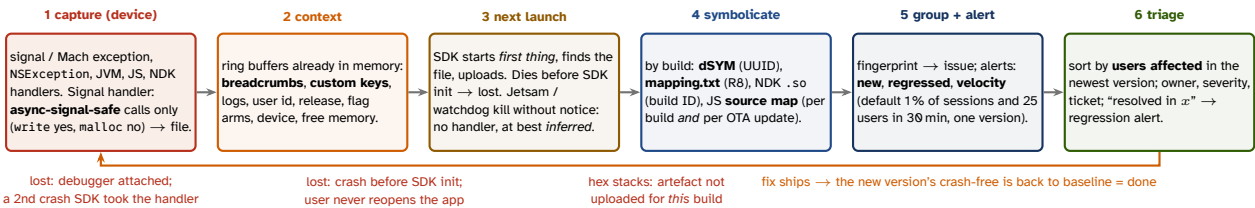


In one line: A crash reporter is a pipeline: an **in-process handler** writes the report at the moment of death, the SDK **uploads it on the next launch**, the server **symbolicates** it with the artefacts *you* uploaded for that exact build, **fingerprints** it into an issue, **alerts** on new, regressed or fast-growing issues, and a human **triages** by users affected. Every stage has a way to silently lose the crash.



The two numbers — and three yardsticks

- Crash-free users** = 1 — crashed users / all users; **crash-free sessions** = 1 — crashed sessions / all sessions. Worked: 1 000 users, 5 000 sessions; 10 users crash 20 times → users **99.0 %**, sessions **99.6 %**. Users = breadth, sessions = frequency; always *per version*, with a minimum sample.

	a user is	a session is	what counts
Crashlytics	one <i>install</i>	cold start, or foreground after ≥ 30 min away	fatal only — not ANRs, not non-fatals
Sentry	a distinct user	launch / foreground; back within 30 s = same one	ends healthy · errored · crashed · unhandled · abnormal (no clean end: battery died)
Play vitals	user × device × day	— (per daily user)	user-perceived (activity shown, foreground service); ANR = input timeouts only

- Same day, 3 sessions, 1 crash: vitals **100 %**, Crashlytics **33 %**. Vitals sees only Play installs on certified devices of users who share data; Sentry sessions are never sampled or filtered (crash events ≠ crashed sessions).
- Sentry **adoption** = a release's share of all sessions (or users) in the last 24 h — read crash-free next to it: at 1% adoption it is a small sample.

Grouping — the fingerprint

- Sentry: **fingerprint** → **stack trace** (in-app frames only: module, normalized filename, context line) → exception type + value → message.
- Splits**: minified JS without source maps, a new wrapper frame, ids or URLs in the message. **Merges**: every network error through one helper.
- Crashlytics: fatals by stack, *variants* = same failure point; logged NSError by **domain + code** — never put ids there.
- Fix: fingerprint rules (`{ default }`) + route, stack-trace rules (wrappers not in-app), merge / unmerge. Sentry's AI grouping only merges *new* issues.

Context that makes a crash fixable

- Breadcrumbs**: screens, requests (method, route, status — no bodies), taps, lifecycle, memory warnings; Crashlytics takes them from Google Analytics.
- Custom keys**: cart size, flag arm, sync state. Crashlytics: **64** pairs (more are not saved), 1kB each; logs **64 kB**, oldest deleted.
- Non-fatals** (decode failure, 5xx): Crashlytics keeps the 8 latest per session. Recording unwinds the stack + writes to disk *synchronously*; on another queue the stack is lost. **User id**: pseudonymous.
- Consent**: `Info.plist` `FirebaseCrashlyticsCollectionEnabled = NO`, then `setCrashlyticsCollectionEnabled(true)` after opt-in (persists across launches); reports stored on the device meanwhile are sent once it is enabled.

Example — enrich, then report

```
let c = Crashlytics.crashlytics()
c.setUserID(account.pseudonymousID) // never email
c.setCustomKeysAndValues(["cart_items": cart.count,
                          "exp_checkout": flags.checkoutArm])
c.log("checkout: pay tapped") // 64 kB ring
do { try await pay() } catch {
  c.record(error: error) // non-fatal
  // Sentry instead: split one call site by route
  SentrySDK.capture(error: error) { scope in
    scope.setFingerprint(["{ default }", "/v1/pay"]) }
```

Alerts and issue states

Crashlytics	fires when	def.
new fatal regressed	a crash or ANR never seen before (new non-fatal: same) a <i>closed</i> issue recurs in a version not yet seen when it was closed	off on
trending velocity	an issue is emerging one issue ≥ 1% of sessions and ≥ 25 users in 30 min, one version; once per issue + version	on off

- Sentry**: *new* = first seen < 7 days. Plain *Resolve*: any later event regresses it; *in next release* (prod 1.8.0, fix merged → only 1.9.0+ reopens). Archive = alerts paused until it escalates. Crash-rate alert: crash-free users or sessions below a set %.
- Route**: page on a new fatal or velocity alert in the newest version; ticket regressed, trending, non-fatals. Slack, Jira, PagerDuty are built in.

Deaths no handler sees

- Watchdog / memory kill** (iOS): no signal. Sentry infers one at next launch if the last run: didn't crash · was in foreground · wasn't the first launch · had no debugger · same app + OS version · wasn't quit by user or OS.
- iOS hang**: a thread pings main (Sentry **2 s**; since 9.0.0 split into fully / non-fully blocking); killed during it → *fatal* hang, sent next launch — Sentry's *app hang rate* counts only these. MetricKit `HangDiagnostic` (iOS 27; `MXHangDiagnostic` deprecated 27.2).
- ANR**: input unhandled **5 s**. RN: a locked JS thread is *not* one — `ScrollView` still scrolls on main.

Symbols — upload in CI, per build

code	artefact	how / trap
Swift/ObjC	dSYM: dwarf-with-dsym in all configs (default: Release only)	run script = last phase; sandboxed → list Input Files
Kotlin/Java	mapping.txt, overwritten every build	Gradle plugin uploads; archive per release
NDK	unstripped .so + GNU build ID	<code>nativeSymbolUploadEnabled</code> + upload task
JS (RN)	source map (Hermes: composed with hermes'c's)	per build <i>and</i> per OTA: <code>sentry-expo-upload-sourcemaps</code>

Which tool

Crashlytics	no-cost; crashes, non-fatals, ANRs (Android 11+); data kept 90 days + release health, hangs, replay; custom grouping; self-host (FSL)
Sentry	no SDK; sharing users + <i>all</i> TestFlight testers; no watchdog, jetsam or code-signature reports
MetricKit	<code>MetricManager</code> (iOS 27; <code>MXMetricManager</code> deprecated 27.2)
Play vitals	per daily user; bad ≥ 1.09 % crash, 0.47 % ANR, 8 % one phone

Run **one** crash SDK — two fight over the handlers (Firebase: Mobile Ads → `disableSDKCrashReporting`).

Traps

- ! Debugger attached — it catches the crash first; detach, relaunch to upload.
- ! 99.9 % overall hiding 100 % for one locale/device — slice first.
- ! Logging *after* a recorded error rotates the useful lines out of 64 kB.
- ! PII in keys, logs, breadcrumbs (emails, tokens, request bodies).
- ! Session replay: Sentry masks all text, images, input by default — unmask only after custom masks; `onErrorSampleRate` keeps a 30 s buffer.